

CO4 - AT3 - COMPARATIVE ANALYSIS

1. Compare Case-Based Learning and Locally Weighted

Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.

1. Case-Based Learning vs Locally Weighted Regression

Practical Example: Recommendation System

Suppose an **online shopping system** recommends products to customers based on their previous purchases, ratings, browsing history, and spending.

Case-Based Learning (CBL)

Case-Based Learning solves a new problem by finding **similar previous cases** and using their solutions.

Example:

- Customer A has purchased mobile phones, earphones, and smartwatches.
- The system searches its database for customers with similar purchase histories.
- If similar customers purchased a particular smartwatch, it recommends that smartwatch to Customer A.
- When a new customer interacts with the system, the new case can be stored for future recommendations.

Locally Weighted Regression (LWR)

LWR predicts an output by giving **higher importance to data points that are closer/similar to the new input**.

Example:

A recommendation system predicts a customer's expected spending based on:

- Number of purchases
- Average rating
- Browsing time
- Previous spending

Nearby/similar customers receive higher weights, while dissimilar customers receive lower weights.

Comparison

Feature	Case-Based Learning	Locally Weighted Regression
Basic idea	Retrieves similar past cases	Fits a local regression model
New data	Compares with stored cases	Uses nearby data points

Feature	Case-Based Learning	Locally Weighted Regression
Adaptability	Very high; new cases can be added easily	High; prediction changes according to local data
Output	Usually retrieves/adapts an existing solution	Produces a numerical prediction
Training	Little or no training required	No global training, but local computation is needed
Computational efficiency	Can be slow with a very large case database	Can be expensive because regression is calculated for each query
Best suited for	Recommendation and classification	Numerical prediction
Example	Recommend products based on similar customers	Predict customer spending

Conclusion

Case-Based Learning is better when recommendations can be made from **similar previous cases**, while **Locally Weighted Regression** is more suitable when the recommendation system needs to **predict a numerical value**. CBL is simple and highly adaptable, whereas LWR provides more mathematical and continuous predictions.

2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.

2. Locally Weighted Regression vs Radial Basis Function Networks

Example Dataset

Consider the following dataset for approximating a function:

x y

1 2.1

2 4.0

3 6.2

4 7.9

5 10.1

6 11.8

7 14.2

x y

8 16.0

The goal is to predict **y** for a new value of **x**, such as **x = 5.5**.

Locally Weighted Regression (LWR)

LWR considers data points **near x = 5.5** more important than distant points.

For example:

- $x = 5 \rightarrow$ high weight
- $x = 6 \rightarrow$ high weight
- $x = 4 \rightarrow$ moderate weight
- $x = 1 \rightarrow$ very low weight

It calculates a local regression using these weighted points.

Main characteristic: Locality depends directly on the query point.

Radial Basis Function (RBF) Network

An RBF network uses fixed **radial basis functions**, usually Gaussian functions, centered at selected points.

For example, centers may be:

- Center 1 = 2
- Center 2 = 4
- Center 3 = 6
- Center 4 = 8

For $x = 5.5$, the centers near 4 and 6 produce stronger responses. These responses are combined to generate the final prediction.

Main characteristic: Locality is determined by the learned/selected RBF centers and their widths.

Comparison

Feature	Locally Weighted Regression	RBF Network
Locality	Based on distance from current query	Based on distance from RBF centers
Model	Builds a local model for each query	Uses learned basis functions
Training	Very little/no global training	Requires training to determine centers and weights
Prediction cost	Can be high for large datasets	Generally faster after training

Feature	Locally Weighted Regression	RBF Network
Memory	Needs access to training data	Stores centers and network parameters
Adaptability	Easily adapts to new data	New data may require retraining/update
Smoothness	Depends on bandwidth/weight function	Usually produces smooth predictions with Gaussian functions
Best use	Small/medium datasets and highly local patterns	Large-scale function approximation
Computational cost	High during prediction	Higher during training, lower during prediction

Locality and Smoothness

LWR:

If the bandwidth is small, only nearby points strongly influence the prediction. This gives a highly local model but may make predictions sensitive to noise.

RBF:

Gaussian basis functions overlap smoothly. Therefore, RBF networks generally provide **smooth function approximation**. Increasing the width of the Gaussian functions makes the output smoother, while smaller widths make it more local.

Computational Efficiency

- **LWR:** Does not require extensive training, but it must search and calculate weights for nearby data whenever a prediction is requested.
- **RBF:** Requires training to find suitable centers and weights, but once trained, predictions are usually computationally efficient.

Final Conclusion

LWR is useful when the relationship changes locally and the dataset is relatively small. It adapts naturally to the query point but can be computationally expensive during prediction. **RBF networks** require more initial training but provide **fast and smooth function approximation**, making them suitable for larger and more complex applications.